

BRACKETS

Minutes 0–15: No cost. The airline's schedule buffer absorbs it. The pdf also states that under 15 is no delay

Minutes 15–60: Only marginal operating costs (crew time, fuel, maintenance) - the data from [16 Cost of delay – EUROCONTROL Standard Inputs for Economic Analyses](#) is used depending on the size of the aircraft to yield cost per minute

- *60 being the top bracket is also backed up by the typical narrowbody turnaround time (same boundary used in the model) beyond this, the cascade to the next leg becomes nearly certain.*

Minutes 60–120: Reactionary delays to other flights start cascading.

At minute 120 EU261 Article 9 kicks in for short-haul flights, triggering mandatory meal and refreshment vouchers

Minutes 120–180: EU261 Article 9 kicks in : meal/refreshment vouchers become mandatory. *the most operationally dangerous bracket because the dispatcher is racing the clock toward the cliff at 180 minutes where it is obligatory to pay out large compensation*

Minutes 180+: EU261 Article 7 kicks in : large fixed compensation payment per passenger. *The lower bound triggers EU261 Article 7 monetary compensation at 3 hours arrival delay, the airline owes €250/€400/€600*

Minutes 300+: *It becomes cheaper to cancel the flight than to continue operating it. EU261 compensation is already owed in full — cancellation adds no marginal Article 7 cost Crew duty time limits are likely breached — forcing an unscheduled crew swap Overnight hotel + transfer costs become unavoidable for many passengers The downstream cascade has fully developed — 4-6 other flights are also delayed*

AIRCRAFT SIZES

SIZE	Distance in km	Num Passengers
S	< 1,500 km	100
M	1,500–5,000 km	150
L	> 5,000 km	250

VARIABLE COST PER MINUTE

Short delays (<30 min) - per-minute cost

SIZE	COST per MIN
S	(15-30) - WE USE 20
M	(40-60) - WE USE 50
L	(110– 180) WE USE 140

FIXED COSTS (EU261)

SMALL

<i>Bracket</i>	<i>Time Interval</i>	<i>Severity</i>	<i>EU261 Cost (EUR)</i>
0	0 - 15 min	On time	0 EUR
1	15 - 60 min	Minor delay	0 EUR
2	60 - 120 min	Moderate delay	500 EUR
3	120 - 180 min	Major delay	500 EUR
4	180 - 300 min	Severe delay	26,000 EUR
5	300+ min	Cancellation zone	26,500 EUR

<i>EU261 trigger reference for this aircraft size:</i>	
<i>Article 9 (meal/refreshment vouchers): triggered at 120 min</i>	<i>~EUR 5-15 per passenger</i>
<i>Article 7 (short-haul, less than 1,500 km): triggered at 180 min</i>	<i>EUR 250 per passenger</i>
<i>Total Article 7 exposure (100 pax)</i>	<i>EUR 25,000</i>

MEDIUM

Bracket	Time Interval	Severity	EU261 Cost (EUR)
0	0 - 15 min	On time	0 EUR
1	15 - 60 min	Minor delay	0 EUR
2	60 - 120 min	Moderate delay	750 EUR
3	120 - 180 min	Major delay	750 EUR
4	180 - 300 min	Severe delay	61,500 EUR
5	300+ min	Cancellation zone	62,250 EUR

EU261 trigger reference for this aircraft size:	
Article 9 (meal/refreshment vouchers): triggered at 120 min	~EUR 5-15 per passenger
Article 7 (medium-haul, 1,500-3,500 km): triggered at 180 min	EUR 400 per passenger
Total Article 7 exposure (150 pax)	EUR 60,000

LARGE

Bracket	Time Interval	Severity	EU261 Cost (EUR)
0	0 - 15 min	On time	0 EUR
1	15 - 60 min	Minor delay	0 EUR

2	60 - 120 min	Moderate delay	0 EUR
3	120 - 180 min	Major delay	0 EUR
4	180 - 300 min	Severe delay	152,500 EUR
5	300+ min	Cancellation zone	153,750 EUR

Using this data below for *hourly* payments, convert all the Table 16.4 hourly rates to per-minute, and find averages for the 3 size classes

Table 16.4: Strategic delay costs per hour - base scenario



Aircraft type	At gate	Taxiing	En-route
A319	€962	€2,281	€4,062
B734	€1,069	€2,566	€4,145
B735	€1,248	€2,756	€4,906
B738	€2,043	€5,036	€8,576
B752	€274	€891	€1,069
B763	€404	€1,164	€1,509
B744	€641	€2,031	€3,802
A319	€712	€2,221	€3,908
A320	€606	€2,031	€3,504
A321	€1,199	€2,459	€4,336
AT43	€1,782	€5,892	€13,007
AT72	€856	€2,839	€5,001
DH8D	€1,556	€4,015	€7,401
E190	€641	€1,354	€1,936
A332	€915	€2,043	€3,267

Source: IATA, 2016; IATA, 2015; IATA, 2014; IATA, 2013; IATA, 2012; IATA, 2011; IATA, 2010; IATA, 2009; IATA, 2008; IATA, 2007; IATA, 2006; IATA, 2005; IATA, 2004; IATA, 2003; IATA, 2002; IATA, 2001; IATA, 2000; IATA, 1999; IATA, 1998; IATA, 1997; IATA, 1996; IATA, 1995; IATA, 1994; IATA, 1993; IATA, 1992; IATA, 1991; IATA, 1990; IATA, 1989; IATA, 1988; IATA, 1987; IATA, 1986; IATA, 1985; IATA, 1984; IATA, 1983; IATA, 1982; IATA, 1981; IATA, 1980; IATA, 1979; IATA, 1978; IATA, 1977; IATA, 1976; IATA, 1975; IATA, 1974; IATA, 1973; IATA, 1972; IATA, 1971; IATA, 1970; IATA, 1969; IATA, 1968; IATA, 1967; IATA, 1966; IATA, 1965; IATA, 1964; IATA, 1963; IATA, 1962; IATA, 1961; IATA, 1960; IATA, 1959; IATA, 1958; IATA, 1957; IATA, 1956; IATA, 1955; IATA, 1954; IATA, 1953; IATA, 1952; IATA, 1951; IATA, 1950; IATA, 1949; IATA, 1948; IATA, 1947; IATA, 1946; IATA, 1945; IATA, 1944; IATA, 1943; IATA, 1942; IATA, 1941; IATA, 1940; IATA, 1939; IATA, 1938; IATA, 1937; IATA, 1936; IATA, 1935; IATA, 1934; 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IATA, 1513; IATA, 1512; IATA, 1511

FINAL PIECEWISE FUNCTIONS (variable + fixed costs):

$$C_{small}(t) = \begin{cases} 0 & \text{if } 0 \leq t < 15 \\ 20t & \text{if } 15 \leq t < 60 \\ 1,200 + 36(t - 60) & \text{if } 60 \leq t < 120 \\ 3,360 + 1,000 + 45(t - 120) & \text{if } 120 \leq t < 180 \\ 7,060 + 25,000 + 54(t - 180) & \text{if } 180 \leq t < 300 \\ 16,640 & \text{if } t \geq 300 \end{cases}$$

$$C_{medium}(t) = \begin{cases} 0 & \text{if } 0 \leq t < 15 \\ 50t & \text{if } 15 \leq t < 60 \\ 3,000 + 55(t - 60) & \text{if } 60 \leq t < 120 \\ 6,300 + 1,500 + 70(t - 120) & \text{if } 120 \leq t < 180 \\ 12,000 + 60,000 + 85(t - 180) & \text{if } 180 \leq t < 300 \\ 25,720 & \text{if } t \geq 300 \end{cases}$$

$$C_{large}(t) = \begin{cases} 0 & \text{if } 0 \leq t < 15 \\ 140t & \text{if } 15 \leq t < 60 \\ 8,400 + 150(t - 60) & \text{if } 60 \leq t < 120 \\ 17,400 + 190(t - 120) & \text{if } 120 \leq t < 240 \\ 40,200 + 153,750 + 230(t - 240) & \text{if } 240 \leq t < 300 \\ 123,900 & \text{if } t \geq 300 \end{cases}$$

These functions model the cost of inaction dependent on the flight delay(t) in minutes, and the size of the aircraft.

FOOT-FORWARD CONTROL EXPENSE OVERVIEW

Bracket 0 - No Delay (<15 minutes)

- No cost incurred.

Bracket 1 - Minor Delay (15-60 minutes)

This is the most common delay band - roughly 90% of all delayed flights according to EUROCONTROL data. The operation can still recover through faster ground tempo without structural intervention. Dispatcher actions in this bracket focus on coordination and resource pre-positioning, not crew or aircraft swap

Action	Cost Range (EUR)	Average (EUR)
Notify gate crew to expedite turnaround	50-150	100
Pre-position fuel, catering, baggage	200-500	350
Check next leg for cascade risk	0-50	25
Log in dispatcher feed	0	0
Total Bracket 1		~EUR 475

Bracket 2 - Moderate Delay (60-120 minutes)

At 60 minutes the operation has lost its slack. The inbound aircraft can no longer make its next departure on time without intervention, and reactionary delays to downstream flights become statistically near-certain. The dispatcher's job shifts from coordination to buffer creation and crew readiness.

Action	Cost Range (EUR)	Average (EUR)
Extend turnaround buffer in schedule	300-800	550
Alert standby crew (notification only, no activation)	100-250	175
Notify connecting flight desks	50-150	100
Verify crew duty time remaining	0-50	25
Total Bracket 2		~EUR 850

Bracket 3 - Major Delay (120-180 minutes)

This is the last bracket in which intervention is cheap. Beyond 180 minutes, the EU261 Article 7 monetary compensation cliff hits, adding tens of thousands of euros to the airline's exposure in a single instant. Bracket 3 is therefore the dispatcher's last clean window to prevent that cliff - meaning interventions here are justified at much higher cost than in earlier brackets.

Note: EU261 Article 9 meal vouchers are now legally owed regardless of intervention. Those costs are handled in the post-delay cost layer and are not included here.

Action	Cost Range (EUR)	Average (EUR)
Activate standby crew (full callout)	1,500-3,000	2,250
Initiate aircraft swap evaluation	200-500	350
Rebook at-risk connecting passengers (proactive)	750-6,000	3,375

Total Bracket 3		~EUR 6,000
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Bracket 4 - Severe Delay (180-300 minutes)

Once past 180 minutes, EU261 Article 7 has already been triggered. The airline now owes monetary compensation to every passenger - EUR 250 to EUR 600 per person depending on flight distance. Foot-forward control at this stage is no longer about preventing the cliff but about containing the damage. The actions become emergency-grade: forced aircraft swaps and unscheduled crew replacement.

Note: The Article 7 compensation payment itself is a post-delay legal cost and is handled in the post-delay layer. The actions below are only the pre-delay control measures the dispatcher can still take to limit downstream cascade and crew duty cliff effects.

Action	Cost Range (EUR)	Average (EUR)
Execute aircraft swap (emergency, mid-day)	5,000-15,000	10,000
Replace crew if duty limit approached	3,000-10,000	6,500
Total Bracket 4		~EUR 16,500

Bracket 5 - Cancellation Zone (300+ minutes)

Beyond 5 hours of accumulated delay, foot-forward control is no longer possible. The decision to cancel becomes a separate post-delay action handled in the cancellation cost layer of the financial engine. There is no pre-delay intervention cost for Bracket 5.

Bracket 5 pre-delay cost: not applicable. All costs are post-delay.

Summary - Foot-Forward Control Costs Across All Brackets

The table below shows the cost of acting early at each bracket. These are the numbers Wingman uses to calculate Expected Avoidable Loss - the metric that ranks the dispatcher's morning queue.

Bracket	Delay range	Action set	Average cost (EUR)	Operational meaning
0	0-15 min	Monitor only	EUR 0	On time - absorbed by schedule buffer
1	15-60 min	4 light coordination actions	~EUR 475	Recoverable through faster turnaround
2	60-120 min	4 buffer + standby actions	~EUR 850	Reactionary delays start propagating
3	120-180 min	4 active interventions	~EUR 6,000	Last clean window before EU261 cliff
4	180-300 min	Emergency swap + crew replacement	~EUR 16,500	EU261 Article 7 compensation triggered

That is the trade-off Wingman exists to make legible - and the foot-forward cost layer is what gives every recommended action a euro figure the dispatcher can defend.

It is important to note that these numbers are significantly lower than those costs incurred with the inaction. That proves us that it is valid and financially relevant to use **Wingman** as it reinforces the foot-forward rather than concurrent control, as it saves an immense amount of money for the airline.

